

Interactive Braille Learning using AI powered Smart Glasses and Computer Vision

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1. Executive Summary:

The proposed project is to develop an AI-powered Braille learning and self-assessment system built on the NUS AiSee smart glasses. Using the glasses' camera, it detects when a learner's finger touches a Braille character, identifies it with a trained computer vision model, and supports two modes: a Learning Mode that reads the character aloud, and a Testing Mode that listens to the learner's spoken response and gives instant feedback. The model is trained on a public Braille dataset and fine-tuned using our own data collected through the glasses under real reading conditions. The expected outcome is a working prototype enabling independent, hands-free Braille practice without requiring constant instructor supervision.

2. Problem Statement:

Braille is the primary reading and writing system for people with visual impairments, but learning it often requires continuous guidance from trained instructors and specialized resources, limiting opportunities for independent practice. Existing solutions mainly focus on Braille recognition, translation, or text-to-speech conversion, while a few educational systems rely on dedicated Braille hardware. However, there is no widely adopted smart glasses-based system that integrates real-time computer vision, interactive learning, and voice-based assessment into a single platform. This project aims to develop an AI-powered smart glasses system that recognizes Braille characters in real time, reads them aloud in **Learning Mode**, and evaluates the learner's spoken answers in **Testing Mode**, enabling accessible, hands-free, and self-directed Braille learning.

3. Data Description:

The project will use publicly available labeled Braille image datasets for training and evaluating computer vision models. The data consists of unstructured image data with Braille character labels. Dataset quality will be maintained through preprocessing, validation, duplicate removal, and class balancing. A data pipeline will support future dataset updates and model retraining. Once an initial model is established, it will be further fine-tuned using data collected through the AiSee smart glasses' own camera under real reading conditions, to improve accuracy and reliability for the intended head-mounted, real-world use case.

Public Datasets:

1. **DSBI (Double-Sided Braille Image Dataset)**
<https://github.com/yeluo1994/DSBI>
2. **Angelina Braille Dataset**
<https://github.com/IlyaOvodov/AngelinaDataset>

3. Braille Character Dataset (Kaggle)

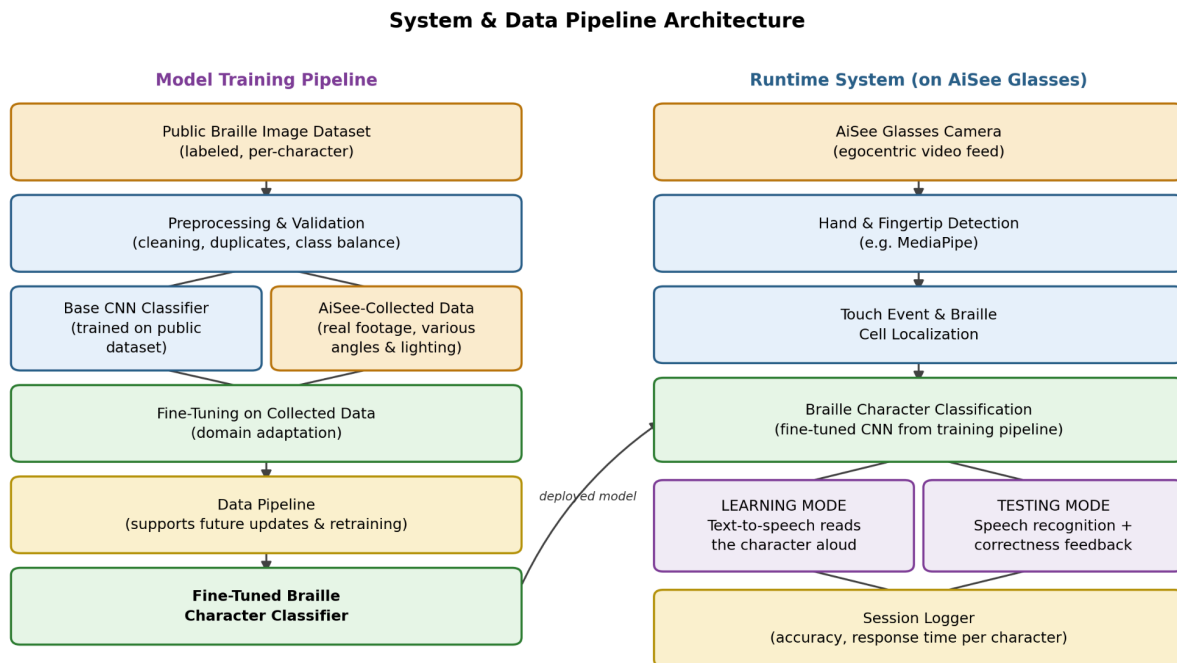
<https://www.kaggle.com/datasets/shanks0465/braille-character-dataset>

4. Tactile Braille Dataset (Zenodo)

<https://zenodo.org/records/7050094>

4. Methods:

The system will use computer vision and deep learning techniques to recognize Braille characters from images captured through smart glasses. The input image will be preprocessed and passed through a trained recognition model to generate the Braille character output. The recognized character will be converted into audio for the Learning Mode, while the Testing Mode will use speech recognition to evaluate the learner's response and provide feedback. The system will integrate image processing, deep learning, speech technologies, and model evaluation techniques.



5. Evaluation Plan

The system will be evaluated using both model-level and system-level metrics. The Braille recognition model will be assessed using precision, recall, F1-score, confusion matrix, and inference time, while speech recognition and feedback modules will be evaluated using response accuracy and error rate. The complete system will be tested for real-time performance, latency, usability, and robustness. Dataset validity will be ensured through proper

annotation verification, class balance checks, train-test separation, and evaluation on unseen samples.

6. Expected Outcomes and Success Criteria:

Expected Outcomes

- A validated Braille image dataset pipeline for training and evaluating AI models.
- A trained computer vision model capable of recognizing Braille characters in real time.
- A smart glasses-integrated application supporting Braille Learning Mode and Testing Mode.
- A complete working prototype demonstrating voice-based guidance and feedback.
- A research paper presenting the methodology, experiments, and results.

Success Criteria

- Achieve reliable Braille character recognition performance on unseen test data.
- Successfully integrate the AI model and application with smart glasses hardware.
- Provide accurate audio output and feedback during learning and assessment scenarios.
- Demonstrate acceptable real-time performance with low latency.
- Complete a research publication/report based on the developed system and evaluation results.

7. Division of work (individual responsibilities)

KURUPPU H.A.

- Data collection, dataset exploration, and data preprocessing pipeline development.
- Perform exploratory data analysis (EDA), data quality validation, and dataset preparation.
- Develop and evaluate Braille recognition models using appropriate deep learning approaches.
- Contribute to model optimization, experimentation, and research documentation.

LAKSHAN H.M.K.

- Data preprocessing, augmentation, and feature preparation for computer vision models.
- Develop computer vision-based Braille detection and classification approaches.
- Perform model training, hyperparameter tuning, and evaluation using research metrics.
- Contribute to smart glasses integration and real-time inference testing.

LAWANYA K.K.H.G.

- Data preparation, analysis, and validation for model development.
- Develop the application workflow integrating the AI model, speech recognition, and text-to-speech components.
- Evaluate system performance, usability, and real-world testing scenarios.
- Contribute to model evaluation, documentation, and research paper preparation.

8. Preliminary Bibliography:

1. Zhong, J., et al. *A Dataset and Benchmark for Braille Image Recognition*. arXiv, 2018.
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<https://arxiv.org/abs/2012.12412>
3. University of Sri Jayewardenepura. *Optical Braille Translator for Sinhala Braille System*.
https://www.researchgate.net/publication/326231538_Optical_Braille_Translator_for_Sinhala_Braille_System_Paper_Communication_Tool_Between_Vision_Impaired_and_Sighted_Persons
4. SLIIT Research. *Machine Translation System to Convert Sinhala and English Braille Documents into Voice*.
<https://rda.sliit.lk/entities/publication/87c430fb-ca3d-40ba-bb90-c7a7b5af9b1e>
5. Angelina Braille Reader – Open-source Braille recognition system.
<https://www.angelina-reader.com/>